



User Manual

PathHunter® CHO-K1 CALCRL-RAMP3 (AM2) β -Arrestin Bioassay Kit

For the Measurement of Ligand-Mediated Arrestin Recruitment

For Bioassay Kits:

93-0252Y2-00219: 2-Plate Kit

93-0252Y2-00220: 10-Plate Kit



Document Number 70-459

Table of Contents

Table of Figures	3
1. Overview.....	4
2. Assay Principle.....	4
3. Materials Provided.....	5
4. Storage Conditions	5
PathHunter CHO-K1 CALCRL-RAMP3 Bioassay Cells.....	5
AssayComplete™ Cell Plating 2 Reagent (CP2)	5
AssayComplete Protein Dilution Buffer (PDB)	6
Control Agonist (Adrenomedullin, rat)	6
PathHunter Detection Kit.....	6
5. Additional Equipment and Materials Required	7
6. Unpacking Cell Cryovials	7
7. Protocol Schematic	9
8. Detailed Protocol	10
8.1 Bioassay Cell Preparation	10
8.2 Agonist Sample Preparation (Rat Adrenomedullin and Test Samples)	11
8.3 Antagonist Sample Preparation	15
8.4 Signal Detection (Preparation and Addition of Detection Reagent)	16
9. Typical Results	17
10. Troubleshooting Guide	18
Document Revision History	19
Limited Use License Agreement	20
Contact Information	20

Table of Figures

Figure 1: Assay Principle.....4

Figure 2: Representative Assay Plate Map14

Figure 3: Typical Results.....17

1. Overview

The PathHunter® CHO-K1 CALCRL-RAMP3 (AM2) bioassay kit provides a robust and highly sensitive functional, cell-based assay to monitor β -arrestin recruitment as a result of ligand-mediated CALCRL-RAMP3 (AM2) activation. The bioassay kit contains all the materials needed for a complete assay, including cryopreserved, ready-to-use cells, detection reagents, cell plating reagent, dilution buffer, positive control agonist, and assay plates. The ready-to-use cryopreserved cells have been manufactured to ensure assay reproducibility and faster implementation from characterization to lot release. This bioassay has been optimized for a 96-well format but can be adapted to a 384-well format if needed.

2. Assay Principle

The PathHunter CHO-K1 CALCRL-RAMP3 (AM2) bioassay monitors activation of the adrenomedullin subtype 2 receptor (AM2) by detecting the interaction of β -Arrestin with the activated GPCR using β -galactosidase (β -gal) enzyme fragment complementation. In this system, CALCRL is fused in frame with the small β -gal fragment called ProLink™ (PK) and co-expressed in cells stably expressing the untagged RAMP3 co-receptor and a fusion protein of β -Arrestin-2 with the larger fragment of β -gal (enzyme acceptor, (EA)). Activation of CALCRL stimulates the recruitment of β -Arrestin-2 to the ProLink-tagged receptor and forces complementation of the two enzyme fragments of β -galactosidase. This results in the formation of an active β -gal enzyme leading to an increase in enzyme activity that can be measured using the chemiluminescent PathHunter Bioassay Detection Reagents. Since arrestin recruitment occurs independent of G-protein coupling, these assays provide a direct, universal platform for measuring receptor activation, thus providing an ideal tool to characterize functional biologics.

β -Arrestin recruitment

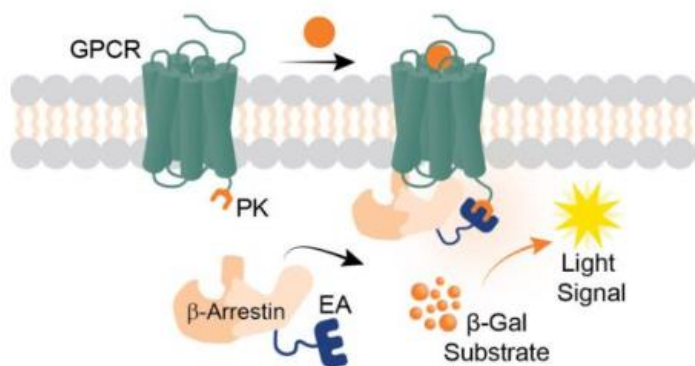


Figure 1. Assay Principle for the PathHunter CHO-K1 CALCRL-RAMP3 (AM2) Bioassay.

3. Materials Provided

List of Components	93-0252Y2-00219 (2-plate Kit)	93-0252Y2-00220 (10-plate kit)
PathHunter CHO-K1 CALCRL-RAMP3 Bioassay Cells (1.2 x 10 ⁶ cells in 0.1 mL per vial)	2 vials	10 vials
AssayComplete™ Cell Plating 2 Reagent (CP2) (100 mL per bottle)	1 x 100 mL	3 x 100 mL
AssayComplete Protein Dilution Buffer (50 mL per bottle)	1 x 50 mL	2 x 50 mL
Control Agonist (Adrenomedullin, rat) *	1 x 500 µg	1 x 500 µg
PathHunter Detection Kit		
Cell Assay Buffer (Bottle)	1 x 11.4 mL	1 x 57 mL
Substrate Reagent 1 (Bottle)	1 x 3 mL	1 x 15 mL
Substrate Reagent 2 (Bottle)	1 x 0.6 mL	1 x 3 mL
96-well White, Clear Flat-bottom, TC-treated Sterile Plates with Lid	2 Plates	10 Plates

* **Note:** The control agonist is used for QC testing of the bioassay cells in this kit, as reflected in the Certificate of Analysis. If needed, additional quantity of the control agonist may be purchased separately (Cat No. 92-1050)

4. Storage Conditions

PathHunter CHO-K1 CALCRL-RAMP3 Bioassay Cells

Cells are shipped on dry ice and should arrive in a frozen state. To ensure maximum cell viability, store the vials of bioassay cells in the vapor phase of liquid nitrogen as soon as possible upon receipt. Please contact technical support immediately, if the cells received were already thawed.

- Short-term (24 hours or less): Store vials at -80°C immediately upon arrival. (DO NOT store at -80°C for longer than 24 hours).
- Long-term (greater than 24 hours): Vials should ONLY be stored in the vapor phase of liquid nitrogen.



Safety Warning: A face shield, gloves and lab coat should be worn at all times when handling frozen vials. Use tongs to remove cryovials from liquid nitrogen storage and place the vials immediately on dry ice in a covered container. Wait at least 1 minute for any liquid nitrogen that may be present inside the vial to evaporate. Do not touch the bottom of the vials at any time to avoid inadvertent thawing of the cells.

AssayComplete™ Cell Plating 2 Reagent (CP2)

Upon receipt, store at -20°C. Once thawed, the Cell Plating 2 Reagent can be stored at 4°C for up to 4 weeks. For longer storage (up to the expiration date listed on the kit's Certificate of Analysis), the reagent should be aliquoted and stored at -20°C. Once aliquoted, do not freeze-thaw more than two

times. Make aliquots suitable for testing one assay plate, for example: 40 mL of reagent per aliquot can be dispensed and stored at -20°C. Ensure that the reagent is equilibrated to room temperature before use in the assay for best performance.

AssayComplete Protein Dilution Buffer (PDB)

Upon receipt, store at -20°C. Once thawed, the Protein Dilution Buffer can be stored at 2-8°C. for up to 4 weeks. For longer storage (up to the expiration date listed in the kit's Certificate of Analysis), the reagent should be aliquoted and stored at -20°C until needed. Avoid multiple freeze-thaw cycles. To make aliquots suitable for testing one assay plate each, 10 mL of reagent per aliquot can be dispensed and frozen down. This may vary depending on your stock sample concentrations and should be adjusted accordingly. Ensure that the reagent is equilibrated to room temperature before use in the assay for best performance.

Control Agonist (Adrenomedullin, rat)

Upon receipt, store at -20°C until ready to use (up to the expiration date listed on the kit's Certificate of Analysis). To avoid condensation, equilibrate the vial to ambient temperature before opening. Centrifuge the vial prior to opening, to maximize recovery, and reconstitute as described on Page 11 of this manual. Once prepared, the stock solution should be stored as suitable aliquots (e.g. 15 µL) at -20°C until needed (avoid multiple freeze thaw cycles). The reconstituted ligand is stable for 12 months at -20 to -80°C or for 1 week at 2-8°C.

PathHunter Detection Kit

Upon receipt, store reagents at -20°C. The detection kit is stable until the expiration date listed on the outer label of the kit box. Thaw reagents at room temperature before use. Once thawed, the reagents can be stored for up to 1 month at 2-8°C. For long-term storage (up to the expiration date listed in the kit's Certificate of Analysis), the reagents should be re-frozen in opaque containers at -20°C, until needed.

For the ten-plate kit: If all the plates will not be used at the same time, we recommend making five aliquots for the Cell Assay Buffer and Substrate Reagent 1. Each aliquot will be adequate for two assay plates. Make five aliquots of 11.4 mL each for Cell Assay Buffer, and five aliquots of 3 mL each for Substrate Reagent 1. Sufficient reagent volumes are provided in the kit to make these aliquots. We do not recommend making aliquots of Substrate Reagent 2, as it does not freeze and is often lost while pipetting

96-Well Clear Bottom Tissue Culture-Treated Plates

Upon receipt, store at room temperature.

5. Additional Equipment and Materials Required

The following equipment and additional materials are required to perform these assays:

Material	Ordering Information
96-Well Green, V-Bottom, Untreated, Non-Sterile Dilution Plates	DiscoverX, Cat No. 92-0011, or similar
Multimode or luminescence plate reader	Refer to the Instrument Compatibility Chart at discoverx.com/instrument-compatibility
Sterile disposable reagent reservoir	Thermo Fisher Scientific, Cat. No. 8094, or similar
50 mL and 15 mL Polypropylene tubes, sterile	
1.5 mL polypropylene microcentrifuge tubes, sterile	
Tissue culture disposable pipettes (1 mL - 25 mL), sterile	
Single and multichannel micropipettes	
Disposable pipet tips for P20, P100, P1000 micropipettes (10 µL-1000 µL)	
Sterile biosafety cabinet	
Humidified tissue culture incubator (37°C, 5% CO ₂)	
Water bath (set to 37°C)	

6. Unpacking Cell Cryovials

Cryovials are shipped on dry ice and contain cells in 0.1 mL of AssayComplete™ Freezing Reagent (refer to the CoA for cell number provided in the vial). Upon receipt, the vials should be transferred to the vapor phase of liquid nitrogen for storage beyond 24 hours. The following procedures are for the safe storage and removal of cryovials from liquid nitrogen storage.



Appropriate personal protective equipment (PPE), including a face shield, gloves, and a lab coat should be worn during these procedures.

1. PathHunter CALCRL-RAMP3 Bioassay Cells must arrive in a frozen state on dry ice.



Contact technical support immediately, if cells received were already thawed.

2. Frozen cells must be transferred to either liquid nitrogen storage or a -80°C freezer immediately upon arrival. If cells will be thawed and used within 24 hours, they can be stored temporarily at -80°C. For long-term storage, place vials in the vapor phase of liquid nitrogen storage.



Cryovials should be stored in the vapor phase of liquid nitrogen ONLY. These vials are not rated for storage in the liquid phase.

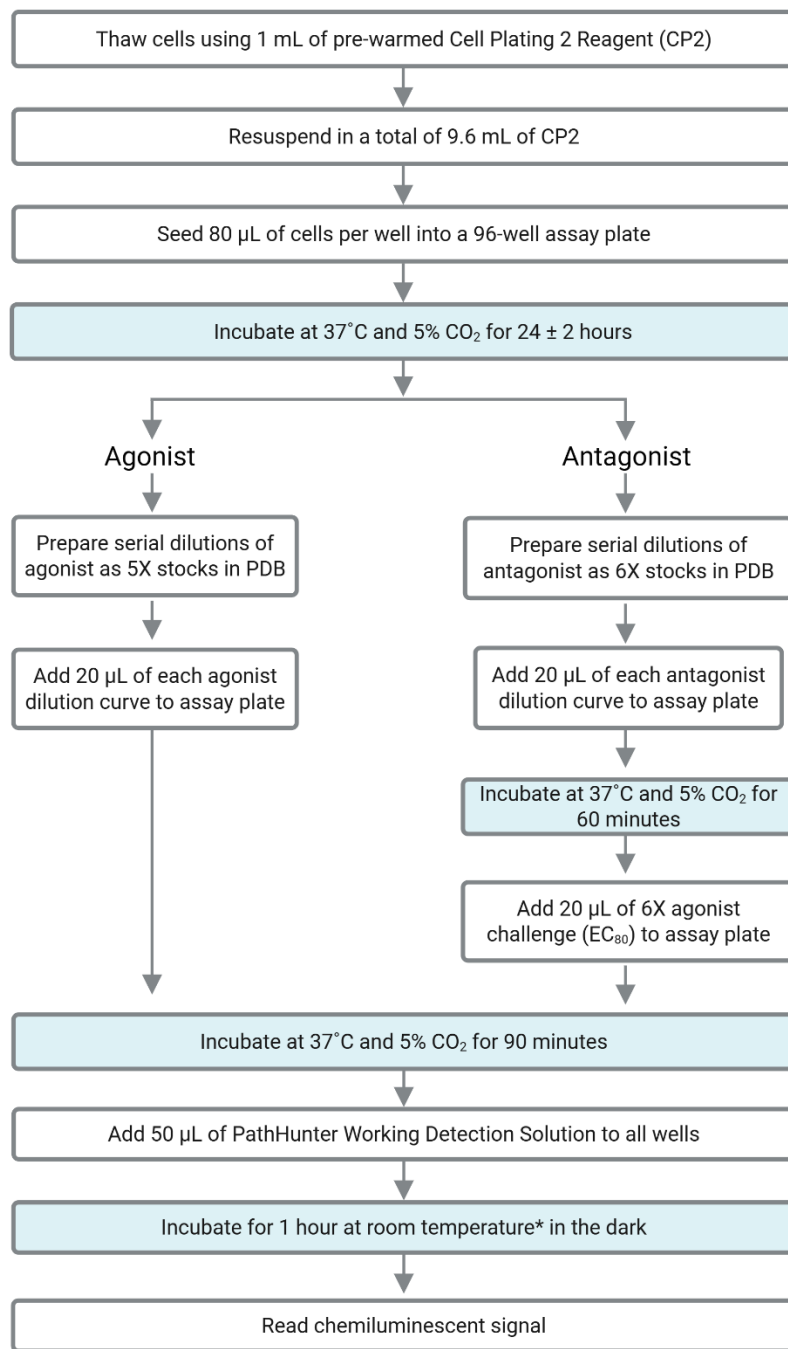
3. Using tongs, remove the cryovials from the liquid nitrogen storage, and place them immediately in

dry ice, in a covered container. Wait for at least one minute for any liquid nitrogen inside the vial to evaporate.

4. Proceed with the thawing protocol in the [Bioassay Cell Preparation](#) section of the [Detailed Assay Protocol](#) section 8.1 below.

7. Protocol Schematic

Quick-Start Procedure: In a 96-well tissue culture treated plate perform the following steps.



*Room temperature refers to a range of 22-25°C

8. Detailed Protocol

This user manual provides a protocol for testing samples in the PathHunter CALCRL-RAMP3 (AM2) Arrestin bioassay in a 96-well format. Prepared volumes may be scaled up or down if required.

This assay is performed under aseptic conditions. Cells and reference standards/test samples are prepared in a Biological Safety Cabinet following good aseptic techniques. All appropriate materials are either certified sterile or prepared aseptically.

8.1 Bioassay Cell Preparation

Day 1

The following protocol is for thawing and plating cryopreserved CHO-K1 CALCRL-RAMP3 bioassay cells from cryovials.

1. Before thawing the cells, ensure that all the materials required are set up in the biosafety cabinet. These include:
 - a. One sterile 25 mL reagent reservoir
 - b. One sterile 15 mL conical tube
 - c. A micropipette (P1000) set to dispense 1 mL
 - d. A multichannel pipette and tips set to dispense 80 µL
 - e. A bottle of AssayComplete™ Cell Plating 2 Reagent (CP2), **pre-warmed in a 37°C water bath for 15 minutes and equilibrated to room temperature.**
 - f. A 96-Well White, Clear-bottom TC-treated Sterile Assay Plate (provided with the kit)
2. Dispense 9.6 mL of CP2 into the 15 mL conical tube.
3. Remove the cryovial from the liquid nitrogen vapor and immediately place it in dry ice.
4. Remove the cryovial from dry ice and ensure cap is tightened. Immediately thaw vial in 37°C water bath for 30 seconds, gently shaking to thaw cells. **DO NOT LEAVE IN WATER BATH.**
5. Visually inspect bottom of vial after 20 seconds. If pellet is thawed, remove vial from water bath, wipe down the outside surface of the cryovial quickly with 70% ethanol, and immediately bring it into the biosafety cabinet. If ice is still visible, return vial to water bath for additional 10-15 seconds.
6. Add 1 mL of pre-warmed CP2 from the 15 mL conical tube into the cryovial. The reagent should be added slowly along the wall of the cryovial tube. Slowly pipette the cell suspension up and down, several times, to uniformly resuspend the cells.
7. Transfer the cell suspension to the conical tube containing the remaining 8.6 mL of CP2. Remove any remaining liquid from the cryovial to ensure maximum recovery of all the cells. Mix by gently pipetting cell suspension up and down 2-3 times with a 10 mL serological pipet.
8. Remove an aliquot (e.g. 20 µL) of cell suspension and count cells using a hemacytometer or automated counter. Adjust cell density to 0.125×10^6 cells/mL with pre-warmed CP2 as needed.
9. Mix cell suspension in conical tube by gentle inversion 2-3 times to ensure that the cells are properly resuspended in the reagent, without creating any froth in the suspension. Immediately transfer the suspension into the sterile 25 mL reagent reservoir.

10. Using a manual 12-channel multichannel pipette, transfer 80 μ L of the cell suspension (10,000 cells per well) to each well of the 96-well assay plate one row at a time, using reverse pipetting. Mix cells in reagent reservoir by pipetting up and down 2-3 times before aspirating and pipetting cells into each subsequent row in the assay plate.
11. Replace the lid and leave the plate at room temperature for 15 minutes, to allow the cells to settle uniformly in the well and minimize potential edge effects.
12. Incubate the assay plate in humidified tissue culture incubator at 37°C and 5% CO₂ for 24 $\pm$ 2 hours before proceeding with the assay.

8.2 Agonist Sample Preparation (Rat Adrenomedullin and Test Samples)

Day 2

The following protocol gives the recommended dilutions for preparing the rat Adrenomedullin (rADM) control agonist and agonist test sample working stocks and serial dilutions.

1. Before beginning, ensure all required reagents are assembled in biosafety cabinet:
 - a. Micropipettes (P20, P100, P1000)
 - b. A multichannel pipette and tips set to dispense 20 μ L.
 - c. An aliquot of AssayComplete™ Protein Dilution Buffer (PDB), pre-warmed in a 37°C water bath for 15 minutes, then equilibrated to room temperature.
 - d. A 96-Well Green, V-Bottom, Untreated, Non-Sterile Dilution Plate, labeled 'Master dilution plate' (MDP)
 - e. Sterile microfuge tubes, for preparation of working stock of Rat Adrenomedullin (rADM).
2. On the day of the assay, prepare rADM working stock solution in PDB as shown in examples in Table 1 below.
 - a. Rat Adrenomedullin (rADM) is supplied as a lyophilized powder. To avoid condensation, equilibrate the vial to ambient temperature before opening. Centrifuge the vial prior to opening, to maximize recovery. Reconstitute by adding 349 μ L of supplied Reconstitution Buffer to the 500 μ g in the vial to generate a 250 μ M stock.
 - i. Reconstituted rADM stock from 2a may be stored in small aliquots (e.g. 15 μ L aliquots) at -20°C.
3. It is recommended to use a minimum aliquot size of 10 μ L for preparation of all sample / working stock dilutions (as shown in Table 1).

Table 1. Example Preparation of Rat Adrenomedullin (rADM) Working Stock

Intermediate and Working Stock Solutions	Final Concentration, μ M	Volume of rADM (Stock concentration)	Volume Protein Dilution Buffer (PDB), μ L
Rat Adrenomedullin (rADM), Working Stock	25	15 μ L (of 250 μ M stock)	135

Note: Use a vortex to mix (10-20 sec) for each dilution before preparing next serial dilution.

4. On the day of the assay, prepare serial dilutions of rat Adrenomedullin (rADM) in row A of the 96-well master dilution plate (MDP), at 5X the final concentration of each dilution, in PDB (i.e. the Ligand Diluent) as per Table 2 below. An example sample dilution scheme is indicated in Table 2 below. Following this dilution scheme should result in sufficient volumes to run triplicate wells per dose in the assay plate.
 - a. Starting at Row A, Well 1 of the MDP, add appropriate volume of ligand diluent (PDB) as indicated in column 6 of Table 2.
 - b. Transfer appropriate volume (indicated in column 5 of Table 2) of rADM Working Stock to the MDP **Row A, Well 2**.
 - c. Prepare the dilution series by transferring the indicated volume of rADM (column 5, Table 2) from the source well (indicated in column 4) to the destination well (indicated in column 1). Pipet up and down several times to mix in destination wells. Replace pipet tips between each serial dilution.
 - d. No rADM working stock is added to well 1 (vehicle only), as this serves as the negative control.

Table 2. Sample concentration in Master Dilution Plate prepared as a 5X stock: Example of Preparation of Rat Adrenomedullin (rADM) Dilutions

Column 1	Column 2	Column 3	Column 4	Column 5	Column 6	Column 7
Well location on Master Dilution Plate (MDP)	Concentration (5X) of rADM on MDP, nM	Dilution Factor	Dilution (5X) rADM Source Well	Volume (5X) of rADM added, μ L	Volume of Ligand Diluent, μ L	Final Concentration (1X) of rADM in Assay Plate, nM
Row A, Well 1	0	--	--	--	100	--
Row A, Well 2	25,000	--	Working Stock (25 μ M; see Table 1)	150	-	5,000
Row A, Well 3	8,333	3	Row A, Well 2	50	100	1,667
Row A, Well 4	2,778	3	Row A, Well 3	50	100	556
Row A, Well 5	926	3	Row A, Well 4	50	100	185
Row A Well 6	309	3	Row A, Well 5	50	100	61.7
Row A, Well 7	103	3	Row A, Well 6	50	100	20.6
Row A, Well 8	34.3	3	Row A, Well 7	50	100	6.9
Row A, Well 9	11.4	3	Row A, Well 8	50	100	2.3
Row A, Well 10	3.8	3	Row A, Well 9	50	100	0.76
Row A, Well 11	1.3	3	Row A, Well 10	50	100	0.25
Row A, Well 12	0.4	3	Row A, Well 11	50	100	0.08

4. On the day of assay, if Test Sample (TS) is an agonist, prepare appropriate dilutions of TS stock as needed, to generate 200 μ L of a working stock of appropriate concentration, that will represent 5X the top desired concentration to be tested in the dose response curve for the TS.

Note 1: Appropriate concentration range and dilution factors for TS should be determined empirically.

Note 2: Depending on stock concentration of TS, intermediate dilutions of TS may need to be prepared (as demonstrated in Table 1 for the rat Adrenomedullin control) to generate the TS working stock, which will be used as the top dose in the dose response curve in Step 5 below.

- a. All dilutions should be prepared in PDB diluent. Volumes may be scaled up or down as needed. However, it is recommended to use a minimum aliquot size of 10 μ L for preparation of all sample / stock dilutions.

- On the day of the assay, prepare serial dilutions of TS in row B of the 96-well master dilution plate (MDP), at 5X the final concentration of each dilution, in PDB (i.e. the Ligand Diluent). Volumes similar to those used in the example in Table 2 will be sufficient to run triplicate wells per dose in the assay plate.

Note: Start top dose of dose-response curve (DRC) in Column 2, as for rat Adrenomedullin (rADM) example in Table 2.

- Transfer 20 μ L of the 5X sample dilutions from the MDP to the appropriate wells containing cells in the assay plate. Refer to example in Figure 2. Representative Assay Plate Map:
 - Row A in MDP: transfer to Rows B, D and F in the assay plate (Rat Adrenomedullin control)
 - Row B in MDP: transfer to Rows C, E and G in the assay plate (Test Sample)
- Replace lid on assay plate and gently mix the contents by slowly moving the plate back and forth 3-4 times on the surface of the biosafety cabinet, in a criss-cross pattern.
- Incubate the assay plate in a humidified incubator at 37°C and 5% CO₂ for 90 $\pm$ 10 minutes.
- Proceed to Signal Detection section (Section 8.4).

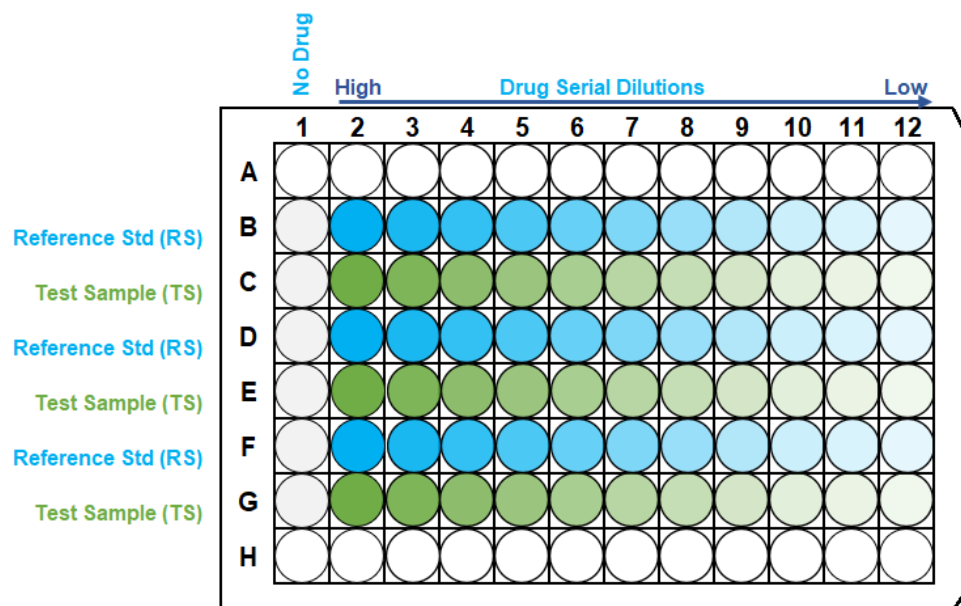


Figure 2. Representative Assay Plate Map

This plate map shows two interdigitated 11-point dose curves, with three replicates per dose point, for a test sample (TS) and control agonist (shown as reference standard in figure). Rows A and H contain cells only but may be used for an assay control with appropriate validation. Column 2 contains the highest dose of each sample, while column 12 contains the lowest dose. Column 1 contains no drug (vehicle only).

8.3 Antagonist Sample Preparation

Day 2:

The following section provides guidance for preparing and evaluating anti-receptor or antagonist test samples. In the below protocol, the test antibody is incubated with the cells for 1 hour at 37°C, followed by stimulation with an EC₈₀ of agonist. However, the optimal pre-incubation time for your antibody/test agent before addition of agonist should be determined empirically.

1. On day of the assay, prepare anti-receptor antibody dilution series at 6X the desired final concentration. Preparation of a 1:3 dilution series of antibody is given below as an example, but the appropriate dilution series for your antibody should be determined empirically. Volumes may be scaled up or down as needed.
 - a. Add 80 µL of Protein Dilution Buffer (PDB) to Columns 1, and 3 -12 of row A of a new master dilution plate (MDP).
 - b. Add 120 µL of 10X anti-receptor antibody sample to **Column 2** of Row A of the MDP.
 - c. Using a clean tip, transfer 40 µL of antibody from Column 2 into Column 3 and mix by pipetting up and down several times. Replace the pipette tip and transfer 40 µL from Column 3 into Column 4. Mix by pipetting up and down several times. Repeat this process until Column 12 is reached, resulting in an eleven point, 1:3 dilution series. No antibody is transferred to Column 1 as this will serve as a negative control.
 - d. Prepare an additional anti-receptor test sample in a similar manner in a new row of the MDP (e.g. Row B).
2. Remove the assay plate containing cells (prepared in Section 8.1) from the 37°C and 5% CO₂ incubator and transfer 20 µL from each row of samples in the MDP to the appropriate wells of the assay plate. See [Representative Assay Plate Map \(Figure 2\)](#) for recommended sample layout.
3. Replace lid on assay plate and gently mix the contents by slowly moving the plate back and forth 3-4 times on the surface of the biosafety cabinet, in a criss-cross pattern.
4. Place the assay plate in the 37°C and 5% CO₂ incubator and incubate for 1 hour.
5. Prepare the agonist challenge.
 - a. Determine the EC₈₀ of rADM and prepare a working stock solution of the agonist challenge at 6X the desired final concentration in PDB.
 - b. Pour the 6X EC₈₀ agonist working stock solution into a sterile reagent reservoir.
6. Retrieve the assay plate from the 37°C and 5% CO₂ incubator and using a multichannel pipet, add 20 µL of 6X agonist challenge to each well of the assay plate.
7. Replace lid on assay plate and gently mix the contents by slowly moving the plate back and forth 3-4 times on the surface of the biosafety cabinet, in a criss-cross pattern.
8. Return the assay plate to the 37°C and 5% CO₂ incubator and incubate for 90 ±10 minutes.
9. Proceed to Signal Detection section (Section 8.4).

8.4 Signal Detection (Preparation and Addition of Detection Reagent)

Day 2:

1. Prepare a working detection solution by mixing the three detection reagent components in a 15 mL conical tube. Refer to the table below for the volume ratios.

Working Detection Solution		
Component	Ratio	Volume per Plate (mL)
Cell Assay Buffer	19	5.7
Substrate Reagent 1	5	1.5
Substrate Reagent 2	1	0.3
Total Volume		7.5

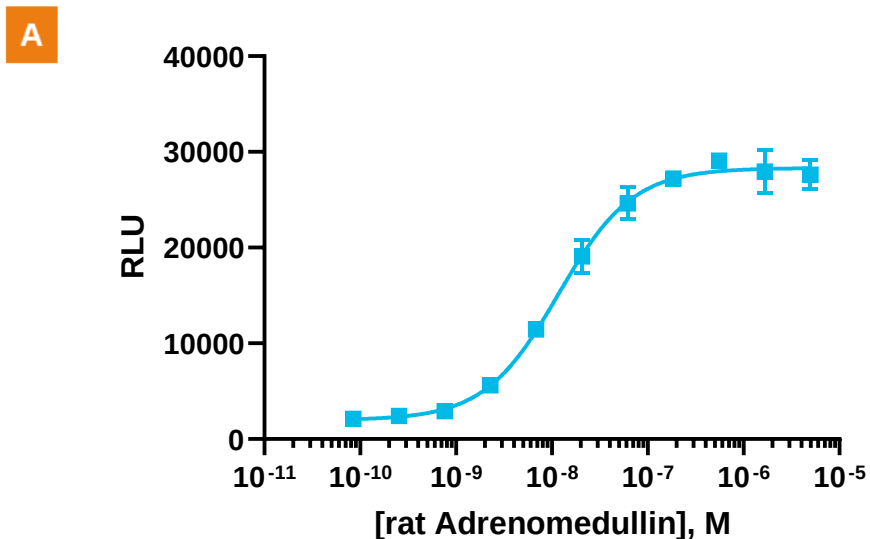
Note: PathHunter Working Detection Solution is stable at room temperature for up to 24 hours with no impact on assay performance

Note: Volume per plate (mL) in table above is recommended volume for a single assay plate. If additional assay plates are to be run in parallel, scale up using volume ratios stated in column 2 of the table.

2. Recap tube and mix contents by gentle inversion; transfer Working Detection Solution into a sterile reservoir.
3. Retrieve assay plate from incubator and remove lid. Pipette 50 µL of the Working Detection Solution from the reservoir into each row of the assay plate.
4. Replace lid on assay plate and gently mix the contents by slowly moving the plate back and forth 3-4 times on the surface of the biosafety cabinet, in a criss-cross pattern. Do not pipette up and down or vortex the plate to mix.
5. Incubate the assay plate for 1 hour (+/- 15 minutes) at room temperature (22-25°C) in the dark (e.g. in a drawer or covered location on the bench top).
6. Read assay plate on a PerkinElmer/Revvity Envision plate reader, or similar instrument, in luminescence mode with 0.2-second/well integration time. Refer to the Instrument Compatibility Chart at discoverx.com/instrument-compatibility/. Data analysis can be performed using any statistical analysis software such as GraphPad Prism, SoftMax Pro, Gen5, or Microsoft Excel etc.

9. Typical Results

The following graph is an example of a typical dose-response curve for the PathHunter CHO-K1 CALCRL-RAMP3 (AM2) Arrestin Bioassay generated using the agonist protocol outlined in this manual. The data shows a potent, dose-dependent increase in β -arrestin recruitment when treated with rat Adrenomedullin. The plate was read on an Envision plate reader (PerkinElmer), with a 0.2 sec/well integration time, and data analysis was conducted using GraphPad Prism.



B

S/B	EC ₅₀ , nM
13	11.7

Typical Results. A. Representative, agonist dose-response curve. B. EC₅₀ and assay window (S/B) for rat Adrenomedullin-mediated CALCRL-RAMP3 (AM2) activation, as measured in this bioassay.

10. Troubleshooting Guide

Problem	Potential Cause	Proposed Solution
No response	Incorrect thawing procedure	Refer to the thawing instructions in the Bioassay Cell Preparation section of this user manual.
	Incorrect ligand used or improper ligand incubation time	Refer to the Certificate of Analysis for recommended ligand and assay conditions.
	Incorrect preparation of the ligand (agonist or antagonist) Sub-optimal time course for induction	Refer to the vendor-specific datasheet to ensure proper handling, dilution, and storage of the ligand. Optimize time course of induction with agonist and antagonist.
Decreased or no response	Cells are not adherent and exhibit incorrect morphology	Confirm adherence of cells using a microscope.
	Problem with microplate reader	The microplate reader should be in luminescence mode. Read at 0.1-1 seconds/well.
Low or no signal	Incorrect preparation of detection reagents	Detection reagents are sensitive to light and should ideally be prepared just prior to use.
	Problem with microplate reader	The microplate reader should be in luminescence mode. Read at 0.1-1 second/well.
Experimental S/B does not match the value noted in the Certificate of Analysis provided	Incorrect incubation temperature	Confirm assay conditions. Check and repeat the assay at the correct incubation temperature, as indicated on the assay datasheet.
	Incorrect preparation of ligand (agonist or antagonist)	Some ligands are difficult to handle. Confirm the final concentration of ligands.
EC ₅₀ is right-shifted	Improper ligand handling or storage	Ensure that the ligands are stored and incubated at the proper temperature.
	Difference in agonist binding affinity	Confirm that the ligand used is comparable to the ligand in the Certificate of Analysis.
	Problems with dynamic range or dilutions	Changing tips during dilution can help in avoiding carryover. Include a cAMP standard curve in the assay to ensure that the ligand dose curve is in the dynamic range of the cAMP detection kit. Moving out of the dynamic range may shift the EC ₅₀ of the ligands.
High variability between replicates	Instrument calibration	Ensure that the dispensing equipment is correctly calibrated, and proper pipetting technique is used.

For questions on using this product, please contact Technical Support at discoverx.com/support/

Document Revision History

Revision Number	Date Released	Revision Details
0	July 2025	New Document

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